

Finite Element Method

Isoparametric Formulation
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- The element displacement function for a two-node element is obtained as

$$u^e(x) = N_1(x^e) u_1^e + N_2(x^e) u_2^e \quad (1)$$

where $x = x_1^e + x^e$ and the shape functions are expressed by

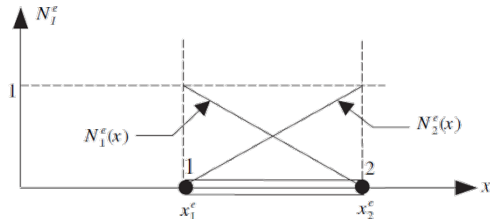
$$N_1(x^e) = 1 - \frac{x^e}{h^e} \quad \text{and} \quad N_2(x^e) = x^e/h^e \quad (2)$$

- Remember that we define a local coordinate system in each element as $x^e = x - x_1^e$ and the element length as $h_e = x_2^e - x_1^e$. If we let

$$\frac{x^e}{h^e} = \xi \quad \text{where } 0 \leq \xi \leq 1 \quad (3)$$

- the shape functions may be rewritten as

$$N_1(\xi) = 1 - \xi \quad \text{and} \quad N_2(\xi) = \xi, \quad 0 \leq \xi \leq 1 \quad (4)$$



- This permits the writing of the element coordinate, displacement, and arbitrary function in the parametric form

$$\begin{aligned}x^e &= N_1(\xi) x_1^e + N_2(\xi) x_2^e \\u^e &= N_1(\xi) u_1^e + N_2(\xi) u_2^e \\w^e &= N_1(\xi) w_1^e + N_2(\xi) w_2^e\end{aligned}\tag{5}$$

- This is called an **isoparametric form**. In this form, the coordinate ξ is a parameter called the **parent coordinate**.
- The advantage of choosing this coordinate system is **1) it is easier to define the shape functions** and **2) the integration over the surface of the element is easier**.
- The computation of x derivatives now requires use of the chain rule which for any function $f(\xi)$ may be written as

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial \xi} \frac{\partial \xi}{\partial x}$$

- If we let $f(\xi) = x$ then we obtain

$$\frac{\partial x}{\partial x} = 1 = \frac{\partial x}{\partial \xi} \frac{\partial \xi}{\partial x}$$

- The term $\partial x / \partial \xi$ is called the **Jacobian**

$$j_e = \frac{\partial x^e}{\partial \xi} = x_2^e - x_1^e = h_e$$

- The derivative of the shape functions may now be computed as

$$\frac{\partial N_a}{\partial x} = \frac{\partial N_a}{\partial \xi} \underbrace{\frac{\partial \xi}{\partial x}}_{j_e^{-1}} = \frac{1}{h_e} \frac{\partial N_a}{\partial \xi}, \quad a = 1, 2 \quad (6)$$

- To satisfy the completeness condition, for any isoparametric shape functions that we may devise, it is necessary to check whether Eq. (7) is satisfied and that the functions are continuous across each point on the element boundary.

$$\sum_a N_a(\xi) = 1 \quad (7)$$

$$N_1(\xi) = 1 - \xi \quad \text{and} \quad N_2(\xi) = \xi, \quad 0 \leq \xi \leq 1$$

- A simple construction for any order we desire is the **Lagrange interpolation formula** given by

$$l_a^p(\xi) = \prod_{\substack{b=1 \\ b \neq a}}^n \frac{(\xi - \xi_b)}{(\xi_a - \xi_b)} = \frac{(\xi - \xi_1)(\xi - \xi_2) \cdots (\xi - \xi_{a-1})(\xi - \xi_{a+1}) \cdots (\xi - \xi_n)}{(\xi_a - \xi_1)(\xi_a - \xi_2) \cdots (\xi_a - \xi_{a-1})(\xi_a - \xi_{a+1}) \cdots (\xi_a - \xi_n)} \quad (8)$$

where the order of the polynomial is $p = n - 1$. In this form we can also pick any range we want for the domain of ξ .

Setting $\xi_1 = -1$ and $\xi_2 = 1$, from Eq. (8) we obtain for $p = 1$ the two shape functions

$$N_1(\xi) = l_1^1(\xi) = \frac{\xi - 1}{-1 - 1} = \frac{1}{2}(1 - \xi) \quad \text{and} \quad N_2(\xi) = l_2^1(\xi) = \frac{\xi + 1}{1 + 1} = \frac{1}{2}(1 + \xi) \quad (9)$$

The Jacobian for this two-node linear element is given by

$$j_e(\xi) = \frac{\partial \xi}{\partial x} = \frac{1}{2}(x_2^e - x_1^e) = \frac{1}{2} h_e \quad (10)$$

The derivatives of the shape functions are given by

$$\frac{\partial N_a}{\partial x} = \frac{2}{h_e} \frac{\partial N_a}{\partial \xi}, \quad a = 1, 2 \quad (11)$$

- For quadratic shape functions we let the nodes be placed at

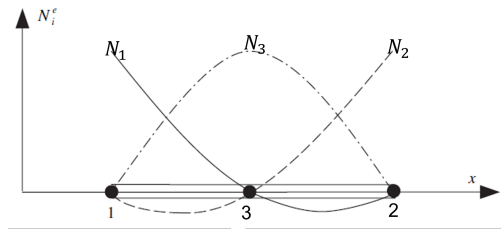
$$\xi_1 = -1, \quad \xi_2 = 1, \quad \text{and} \quad \xi_3 = 0$$

- and obtain the three shape functions as

$$\begin{aligned} N_1(\xi) &= l_1^2(\xi) = \frac{(\xi - 1)(\xi - 0)}{(-1 - 1)(-1 - 0)} = \frac{1}{2}\xi(\xi - 1) \\ N_2(\xi) &= l_2^2(\xi) = \frac{(\xi + 1)(\xi - 0)}{(1 + 1)(1 - 0)} = \frac{1}{2}\xi(\xi + 1) \\ N_3(\xi) &= l_3^2(\xi) = \frac{(\xi + 1)(\xi - 1)}{(0 + 1)(0 - 1)} = 1 - \xi^2 \end{aligned} \quad (12)$$

- Note we have placed the local node 3 in the interior of the element. It is sometimes convenient to let node 1 and node 2 be the boundary nodes of the element with any interior nodes numbered as node 3 and higher.
- If we let the coordinates for the element be given by x_1^e , x_2^e , x_3^e , the parametric form for the physical element is given by

$$x = N_1(\xi) x_1^e + N_2(\xi) x_2^e + N_3(\xi) x_3^e \quad (13)$$



Mapping of the one-dimensional domain from the parent domain $[-1, 1]$ to the physical domain $[a, b]$.

- The Jacobian is now given by

$$j_e(\xi) = \frac{\partial x}{\partial \xi} = \left(\xi - \frac{1}{2}\right) x_1^e + \left(\xi + \frac{1}{2}\right) x_2^e - 2\xi x_3^e = \frac{1}{2} h_e + \xi(x_1^e + x_2^e - 2x_3^e) \quad (14)$$

- Note that the Jacobian is not constant unless the coordinate for node 3 is placed at the middle of the element.

$$x_3^e - x_1^e = \alpha (x_2^e - x_1^e), \quad \frac{1}{4} < \alpha < \frac{3}{4}$$

- Furthermore, for a valid isoparametric mapping, the Jacobian must be positive for all values of ξ . This restricts the location for the center node to satisfy

$$\frac{\partial N_a}{\partial x} = \frac{1}{j_e(\xi)} \frac{\partial N_a}{\partial \xi}, \quad a = 1, 2, \dots, n \quad (15)$$

- A general form for the one-dimensional shape function derivative is given by the rational expression.

- Since the shape functions are expressed in terms of the parent coordinate, it also is convenient to express the integrals in terms of the parent coordinate.
- Note that the differential is given in terms of the Jacobian by

$$dx = \frac{\partial x}{\partial \xi} d\xi = j_e(\xi) d\xi \quad (16)$$

- Thus, our integrals may be expressed as

$$\int_0^{h_e} f(x) dx = \int_{-1}^1 \hat{f}(\xi) j_e(\xi) d\xi \quad (17)$$

where $\hat{f}(\xi) = f(x(\xi))$.

- We use **quadrature** (numerical integration) to approximate the integrals. The quadrature method offers a sufficiently accurate integration for use in a finite element calculation. Therefore, the integral is replaced by the form

$$I = \int_{-1}^1 f(\xi) d\xi \approx \sum_{j=1}^n f(\xi_j) w_j \quad (18)$$

- The number of integration points needed to integrate a polynomial of order p exactly is given by

$$n_{\text{gp}} \geq \frac{p+1}{2} \quad (19)$$

Table Position of Gauss points and corresponding weights.		
n_{gp}	Location, ξ_i	Weights, W_i
1	0.0	2.0
2	$\pm 1/\sqrt{3} = \pm 0.5773502692$	1.0
3	± 0.7745966692	0.555 555 5556
	0.0	0.888 888 8889
4	± 0.8611363116	0.347 854 8451
	± 0.3399810436	0.652 145 1549
5	± 0.9061798459	0.236 926 8851
	± 0.5384693101	0.478 628 6705
	0.0	0.568 888 8889
6	± 0.9324695142	0.171 324 4924
	± 0.6612093865	0.360 761 5730
	± 0.2386191861	0.467 913 9346

- Evaluate the integral below using two-point Gauss quadrature.

$$I = \int_2^5 (x^3 + x^2) dx$$

- Using Eq. (19), the minimum number of Gauss points is obtained as

$$2n_{\text{gp}} - 1 = 3 \Rightarrow n_{\text{gp}} = 2$$

- From the Table on the previous slide, positions of Gauss points and the corresponding weights are

$$W_1 = W_2 = 1, \quad \xi_1 = -\frac{1}{\sqrt{3}}, \quad \xi_2 = \frac{1}{\sqrt{3}}$$

- Using Eqs. (5) and (9) $x = 3.5 + 1.5\xi$. Also,

$$f(x) = x^3 + x^2, \quad \hat{f}(\xi) = f(x(\xi)) = (3.5 + 1.5\xi)^3 + (3.5 + 1.5\xi)^2$$

- From Eq. (10)

$$j_e = \frac{1}{2} h_e = \frac{1}{2}(5 - 2) = 1.5$$

- Therefore, using Eqs. (17) and (18)

$$\begin{aligned} I &= \int_{-1}^1 1.5((3.5 + 1.5\xi)^3 + (3.5 + 1.5\xi)^2) d\xi \\ &= 1.5((3.5 + 1.5\xi_1)^3 + (3.5 + 1.5\xi_1)^2)W_1 + 1.5((3.5 + 1.5\xi_2)^3 + (3.5 + 1.5\xi_2)^2)W_2 = 191.25 \end{aligned}$$

- In this case, Gauss quadrature is exact. You can check the result by performing analytical integration.

Next: Divergence Theorem

Thanks for your attention!